

Binomial & Poisson $\left\{ \begin{array}{l} \text{Discrete} \\ \text{Count} \end{array} \right\}$ & $\left\{ \begin{array}{l} \text{continuous} \\ \text{time dependent} \end{array} \right\}$ distribution

PAS

Normal

$$\text{std} = \sqrt{npq}$$

$$\text{mean} = np$$

$$\text{var} = npq$$

Binomial distribution

$$\left[{}^nC_r = \frac{n!}{(n-r)!r!} \right]$$

$$P(x) = \left(\frac{n}{x} \right) p^x q^{n-x}$$

$$P(x) = \frac{n!}{(n-x)!x!} p^x q^{n-x}$$

Q. Coin tossed 5 times, using B.D.

a) Exactly 2 heads

$$P(x=2) = \frac{5!}{3!2!} \times (0.5)^2 (0.5)^3$$

$$\Rightarrow \frac{5 \times 4 \times 3!}{3! \times 2!} \times \frac{1}{2^5}$$

$$\Rightarrow \frac{10}{32} \rightarrow \boxed{\frac{5}{16}}$$

b) At least 4 heads

$$P(x=4) + P(x=5)$$

$$\frac{5!}{4!1!} \times (0.5)^4 + \frac{5!}{5!0!} \times (0.5)^5$$

$$\Rightarrow \frac{5 \times 4!}{4!} \times (0.5)^4 + (0.5)^5$$

$$\Rightarrow \frac{5 \times 4!}{4!} \times (0.5)^4 + (0.5)^5 \rightarrow \frac{6 \times 1}{32}$$

$$\rightarrow \boxed{\frac{3}{16}}$$

When we know event = toss, No. of times event repeated, find probability, then we use Binomial distribution

At least one head

$$c) P(x \geq 1)$$

$$P(x \geq 1) =$$

$$P(1) + P(2) + P(3) + P(4) + P(5)$$

excluding all Tail.

$$\Rightarrow \frac{31}{32}$$

$\frac{9!}{6!3!} \left(\frac{4}{5}\right)^6 \left(\frac{1}{5}\right)^3$
 $p = 0.8$
 $n = 9$
 $x = 6$
 $\frac{39 \times 817}{8 \times 2} \times \frac{4^6}{5^9}$

Normal distribution:-

Continuous distribution. $\rightarrow$ depends on earlier events.

$\Rightarrow \frac{3,44,064}{1953125}$
 $\Rightarrow 0.176$

The

$$n(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi \times \sigma^2}} e^{-\frac{1}{2\sigma^2} (x-\mu)^2}$$

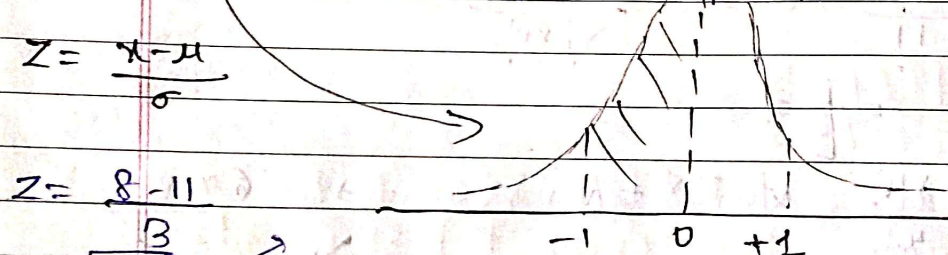
$$= \frac{1}{\sqrt{2\pi} \sigma} e^{-\frac{1}{2\sigma^2} (x-\mu)^2}$$

$[e = 2.7189]$
 $[\pi = 3.14]$

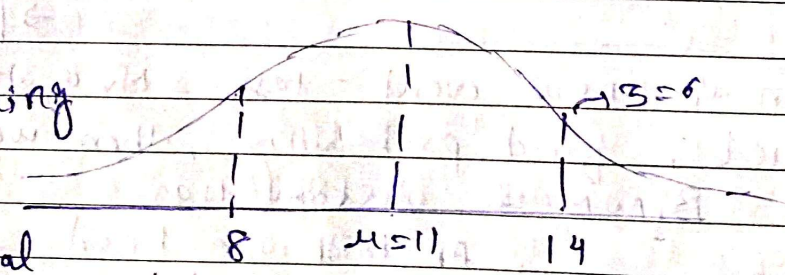
Area under curve = 1

Std Normal distribution:-

The distribution of Normal random variable with mean = 0 & variance 1 is called Std Normal distribution.



Transforming into Std Normal distribution



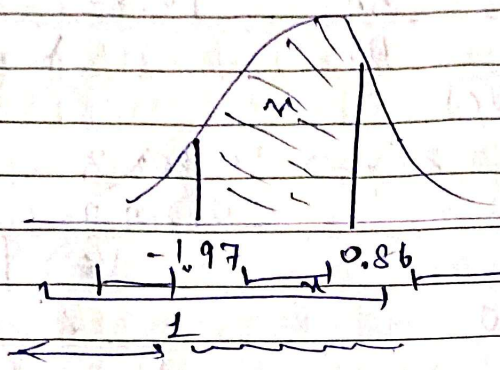
$Z \rightarrow$ gives the left area of the ~~Area~~ curve

Q) Find AUC.

a) to the right of $z = 1.84$

$1 - \text{Area}(1.84)$
~~1 - 0.9699~~
 right

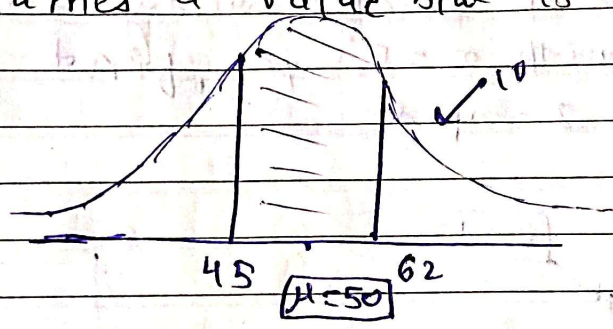
b) Now $z = -1.97$ & $z = 0.86$



Area
 $(-1.97 = 0.2442)$
 $0.86 = 0.8051$

$1 - 0.2442 + 0.8051 = 1$

Q) Given a random variable x having a normal distribution with $\mu = 50$ & $\sigma = 10$, find probab that x assumes a value b/w 45 & 62.

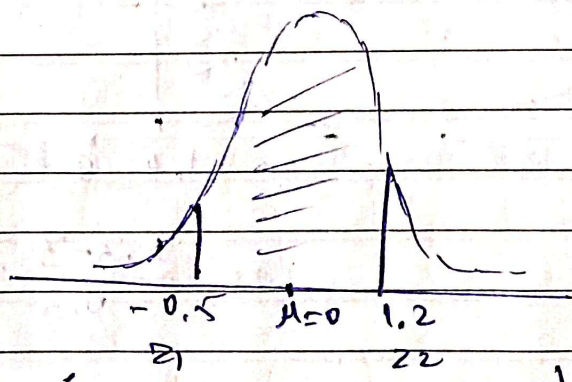


Standard form

$z_1 = \frac{x - \mu}{\sigma}$

$z_1 = \frac{45 - 50}{10} = -0.5$

$z_2 = \frac{62 - 50}{10} = 1.2$



$(0.3085 \quad 0.8844)$

$\left[\begin{array}{c} 0.8844 \\ -0.3085 \end{array} \right]$
 (Ans)

(c)

$\mu = 800 \text{ hrs}$ $\sigma = 40$

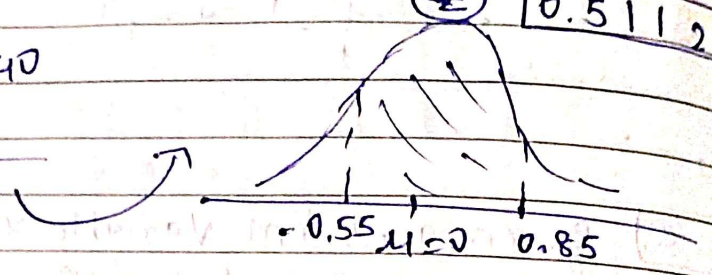
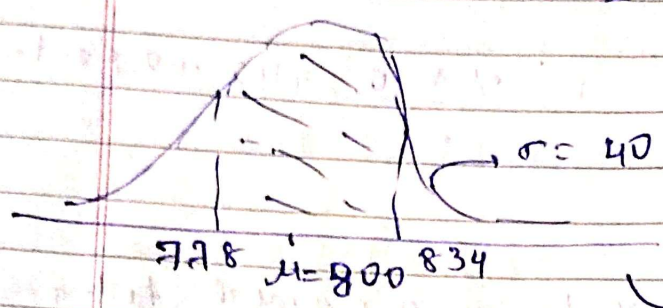
$x_1 = 778$
 $x_2 = 834$

$z_1 = \frac{778 - 800}{40} \Rightarrow \frac{-22}{40} = -0.55$ $\rightarrow 0.2911$

$z_2 = \frac{834 - 800}{40} \Rightarrow \frac{34}{40} = 0.85$ $\rightarrow 0.8023$

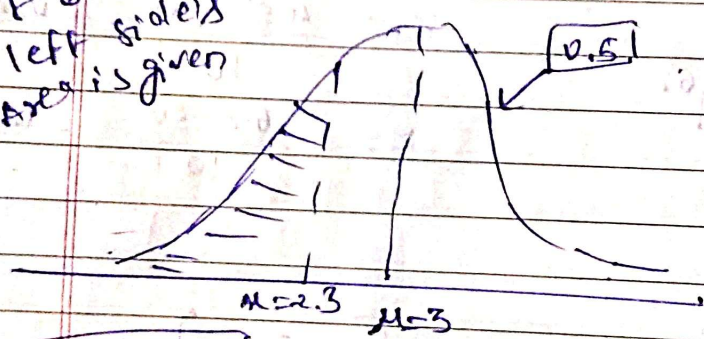
(AUC)

$\therefore z_2 - z_1 \rightarrow 0.8023 - 0.2911 = 0.5112$

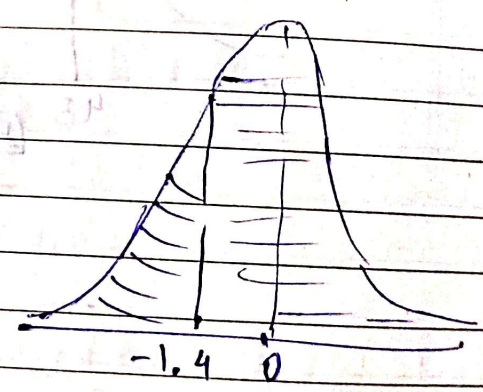


Q1) A certain type of storage battery lasts, on avg 3.0 yrs, with $\sigma = 0.5 \text{ yr}$, find prob at least 2.3 yrs.

Z-table me
left side's
Area is given



0.0808
Z-table
value.
 $1 - 0.0808 = 0.9192$



$\frac{2.3 - 3}{0.5} = -1.4$

Poisson Distribution:-

→ Predicts the probability of certain events happening when you know how often event has occurred.

∴ t = How much time
 λ → Avg.

$$P(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!}, \quad x = 0, 1, 2, \dots$$

λ → average number of outcomes per unit time,

Q → The avg no. of major storms in city is 2 per year, what is probab that exactly 3 storms will hit your city next year?

$$P(x; \lambda t) = \frac{e^{-\lambda t} (\lambda t)^x}{x!} \quad \begin{array}{l} \lambda = 2 \\ x = 3 \\ t = 1 \text{ yr} \end{array}$$

$$\Rightarrow \frac{(2.7)^{-2} (2 \times 1)^3}{3!} \rightarrow \frac{0.137 \times 8}{6} \\ \Rightarrow \frac{0.548}{3} \times 0.182$$

Q → Not requested at all $x=0$
↳ $\lambda=5$

$$\frac{e^{-\lambda t} (\lambda t)^x}{x!} \quad t=1 \\ \text{∴ } \frac{(2.7)^{-5} \times (1)}{1} \rightarrow 0.0069$$